cloudtorch / develop — Step 3a outline

Composite forward model on a *pre-calculated, fixed* background ⊕ two clouds — the given-background **sibling** of step 3, so background handling becomes a user-selectable mode (*fit* vs *given*). Per-pixel, strategy-agnostic, differentiable, no regularisation. All PyTorch; **no fitting here**.

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1. OBJECTIVE

Provide the *alternative* to step 3's fitted PCA background: a composite forward model that stacks the existing two clouds on a background that is **supplied, not fitted** — the pre-calculated skglm quiet-Sun profile described in `cloudtorch_summary` (`bdata` = `new_fit` − `new_extrafit`), one fixed spectrum per pixel *and* per frame. This re-instates the *original* cloudtorch approach inside the modular `develop/` stack and makes the background a **mode switch**: step 3 (*fit*, 6 PCA coeffs/pixel) ↔ step 3a (*given*, 0 background DoF). Same atomic unit — one pixel → one spectrum — differentiable now w.r.t. the cloud parameters *only*.

2. THE GIVEN BACKGROUND (WHAT IT IS, HOW IT ALIGNS)

$I_{\rm in}$ = `bdata` = `new_fit` − `new_extrafit` from `mihi_rvm_reconstruction_averages_skglm.npz` (shape (t, y, x, λ), same as `data`): the skglm fit with its *extra* (filament) absorption removed, i.e. the incident intensity the cloud model needs — *no* PCA, *no* free coefficients. **Live trap (now load-bearing):** the given background must be carved with the *identical* border+λ crop (`[100:500]`), the *identical* fit window (`ts,ys,xs`), and divided by the *identical* `Iqs` as the observed data — reuse the run's `Iqs` from `load_data` (the red-continuum scalar), never a re-derived one, or the transfer $I_{\rm out} = I_{\rm in}e^{-\tau} + S(1 - e^{-\tau})$ sits at the wrong absolute level. Memory-map the (∼11.6 GB) file and slice; keep only the fit-window block on device.

3. COMPOSITE FORWARD MODEL (GIVEN)

$$I_{\rm out} = \mathrm{cloud}_2\big(\mathrm{cloud}_1(I_{\rm in})\big), \qquad I_{\rm in} = \texttt{bdata}_{\rm pix}\ \text{(fixed)}.$$

i.e. `cmt.model_synth_2clouds_givenbck(x, p_cloud, I_in)` with $I_{\rm in}$ a *precomputed constant* tensor — the `background(pca,c)` call is simply dropped. Implementation: refactor the shared tail of step-3 `composite_synth` into a private `_compose(x, I_in, p_cloud, loga_clamp)`; step 3's PCA path calls it after `background()`, step 3a calls it directly. **Step 3 stays byte-for-byte**; 3a adds a sibling entry point `composite_synth_given(x, I_in, p_cloud, ...)`. Per-pixel independence is unchanged (a row is one pixel now, one (x, y, t) sample later).

4. PARAMETERS — AND WHY THIS MODE IS BETTER-POSED

Per sample: **0 background** + **8 cloud** free ($S, \Delta\tau, v_{\rm los}, \Delta v$ per cloud; `loga` frozen → 10 columns), vs 6+8 in step 3. Because the background is *fixed*, the background/cloud **degeneracy is gone by construction**: the background cannot grow the emission spikes (or fill its Hα core) that the clouds then cancel — the pathology steps 5–6 needed the `no_emission` cap and `background_anchor` to suppress *cannot arise here*. So the *given* mode is intrinsically well-posed and needs only the *light cloud* guardrails ($S, \Delta\tau \geq 0$; $\Delta v \geq 8$; $|v_{\rm los}| \leq v_{\rm max}$); the background terms are simply switched off.

5. LOSS, STRATEGY-AGNOSTIC, AND THE TOGGLE

Reuse `chi2_per_pixel` *unchanged* (per-pixel vector + scalar mean); no `regu*` here. Same outer-layer choice as step 3 (classical *or* PINN); note for later — in *given* mode a PINN field predicts only the **8 cloud** numbers (a smaller net), but that is step 4a, not now. The **mode switch** `background_mode`∈`{fit,given}` is the integration hook: step 3a delivers the *given* branch — `composite_synth_given` + a light `load_given_background(cfg, meta)` helper returning the aligned, `Iqs`-normalised (N, L) tensor. Wiring the toggle through the PINN and `fit_cube.py` is deferred (steps 4a / 6-integration).

6. DECISIONS (PROPOSED) & ONE THING TO CONFIRM

- **a** **Given background** = `new_fit` − `new_extrafit` (per `cloudtorch_summary`); used as $I_{\rm in}$ as-is.
- **b** **Strictly fixed** — 0 background DoF (default). An optional global scale (or low-order residual correction) on $I_{\rm in}$ stays a *deferred lever* if it proves slightly mis-normalised; *not* in 3a.
- **c** File path: the reconstruction `.npz` sits in the *same directory* as the data cube, so `data.bkg_path` **defaults to the resolved data path with the filename swapped** (`mihi_rvm_reconstruction_averages_skglm.npz`); a `-set data.bkg_path=...` override remains. Read header-only / sliced, never fully loaded.
- **d** **Confirmed present** on the server (same directory as the data cube) and, per `cloudtorch_summary`, the same (t, y, x, λ) shape — so it supplies $I_{\rm in}$ for every fitted frame `ts`.

After your review: refactor `composite_model.py` (`_compose` + `composite_synth_given`), add `load_given_background`, and a step-3a test cell (load `bdata`, align, confirm grads reach only the cloud params, smoke-fit, obs-vs-fit panels, and compare χ^2 against the step-3 fitted background). Then integrate `background_mode` across steps 4 and 6.